Empirical Analysis of Mean Reversion in USDT/USD

Theory Alignment, Estimation, and Out-of-Sample Evidence

Empirical appendix for `meanreversion_article_main.tex`

9 August 2026

Abstract

This document implements and audits the stablecoin application of the copula-based mean-reverting processes developed in `meanreversion_article_main.tex`. The core parametric mixed-OU (MOU) and mixed-CIR (MCIR) transition kernels are exactly the constant-intensity Poisson-switching special cases of the article's mixed-copula construction. The affine transformed OU and CIR specifications are restricted versions of the article's linear transformation solutions. The nonparametric, threshold, rolling, and forecasting exercises are empirical diagnostics that are not claimed as additional theoretical propositions.

Using 524,610 exact five-minute transition pairs from Kraken USDT/USD over 2021–2025, the estimated free-mean OU has a long-run log-price deviation of 0.665 basis points and a half-life of 589.5 minutes. The jointly estimated MOU substantially improves the in-sample transition likelihood. A matched-design, jointly re-estimated boundary bootstrap rejects the OU restriction (LR = 12,722.80, B = 500, corrected p = 0.001996), although neither model passes the residual specification tests and MOUF does not dominate OUF in the original forecast exercise. A diagnostic two-scale mixed copula attains the highest out-of-sample copula log score, while the two-scale Gaussian copula most closely matches the empirical dependence curve. These two-scale specifications are not one-dimensional Markov models and the empirical marginal is not verified to satisfy the article's mean-reversion sufficient condition. The evidence supports the mixed kernel as a useful density extension, but does not identify its reset intensity with a unique economic mechanism. A second-stage registry estimates or explicitly diagnoses all 25 finite-dimensional article and methodology objects before ranking them. On the untouched 2024 selection sample, an exact filtered two-state Markov-switching OU has the highest five-minute conditional log score (8.82469), ahead of a Student-t marginal and Student-t copula (8.78120). The day-block interval for the difference is strictly positive, but both models have weakly identified structural parameters. Model emphasis must therefore distinguish predictive performance from structural identification.

1 Relationship to the theoretical article

The article defines a Markov process as mean reverting when it has an asymptotic mean and its infinitesimal conditional drift has the sign of the displacement toward that mean. It then constructs new processes through (i) monotone transformations of OU and CIR diffusions and (ii) mixtures of a consistent base copula family with the independence copula. Table 1 records the exact relationship between those constructions and the empirical implementation.

The article does not specify an estimator in its currently empty empirical section. Maximum likelihood, composite likelihood, block bootstrap, PIT diagnostics, and rolling out-of-sample scores are therefore empirical additions; they do not alter the theoretical process definitions.

Table 1: Theory-to-implementation alignment

Article object	Empirical implementation	Audit verdict
Mean-reversion definition	Five-minute local-linear estimate of $\mu_h(x) = h^{-1}\mathbb{E}[X_{t+h} - X_t \mid X_t = x]$ and sign statistic $\mathbf{1}\{(\theta - x)\mu_h(x) > 0\}$	Finite-horizon proxy; it does not prove the $h \downarrow 0$ limit or asymptotic moment convergence.
Stationary OU and CIR	Exact OU transition density for $X_t = \log P_t$; CIR transition density for P_t and for absolute depeg pressure	Model equations match. Price CIR is numerically weakly identified.
Linear transformed OU	$X_t = A_0 e^{(\kappa-d)t} r_t + B_0 e^{-dt}$ with $A_0 = 1$ and $B_0 = 0$	Exact restricted special case of the article's linear OU solution.
Linear transformed CIR	$P_t = A_0 e^{(\kappa-d)t} r_t + \theta[1 - A_0 e^{(\kappa-d)t}] + B_0 e^{-dt}$, with $A_0 = 1$ and $B_0 = 0$	Exact restricted formula; base CIR parameters are held fixed and the resulting model is weakly identified.
Mixed copula $C^{\mathrm{mix}}_{s,t} = a_{s,t} C_{s,t} + (1 - a_{s,t})\Pi$	MOU and MCIR with $a_{s,t} = e^{-\lambda(t-s)}$ and the stationary OU/Gamma marginal in the reset component	Exact constant-intensity case of the article's Poisson construction.
General $a_{s,t} = \exp\{-\int_s^t c(u)\,\mathrm{d}u\}$	Constant $c(u) = \lambda$ only	Time-varying reset intensity is not estimated.
Gaussian copula with a general marginal	Gaussian and mixed Gaussian copulas with an empirical marginal	Copula construction is consistent, but the article's sufficient sign condition for mean reversion is not imposed and is not supported by the fitted finite-horizon drift.
Two-scale latent Gaussian copula	A weighted sum of fast and slow OU covariance components, optionally combined with an independence reset	Diagnostic extension: the two-dimensional latent state is Markov, but the observed scalar state is generally not one-dimensional Markov and is not an article proposition.
Nonlinear OU/CIR transformations	Exponential, quadratic, and special-function solutions are not estimated	Outside the present empirical scope.
Mixed-copula bond-pricing PDE	Not used for stablecoin prices	Not applicable to the stated empirical objective.

The article's line following its diffusion equation states that $\alpha(t)$ is a positive constant and then writes $\mu(x) = \alpha(x - \theta)$. Under the article's own sign definition this is mean repulsion. Consistency requires either $\alpha < 0$ or $\mu(x) = \alpha(\theta - x)$ for $\alpha > 0$. The empirical code uses the latter, mean-reverting convention throughout.

2 Data, state variables, and exact transition design

The raw file is Kraken's five-minute USDT/USD OHLCVT series. Its SHA-256 digest is

`417ea3aa71f86c4476c30568f251f3f873ae8b0d44c5f480f5ad2f35fc6acc22`.

The audit finds 797,178 observations from 29 March 2017 to 31 December 2025, 750,220 exact 300-second adjacent pairs, no duplicate timestamps, no missing fields, and grid coverage of 86.5186%. The main sample is 2021–2025. The training, validation, and test periods are 2021–2023, 2024, and 2025, respectively.

Table 2: Sample construction

Sample	Period	Observations	Coverage (%)	Exact 5-minute pairs
Baseline	2021–2025	524,835	99.7998	524,610
Extended	2020–2025	627,782	99.4434	625,690
Full	2017–2025	797,178	86.5186	750,220
Training	2021–2023	314,759	99.8094	314,590
Validation	2024	105,153	99.7581	105,118
Test	2025	104,923	99.8126	104,900

Source file: `output/tables/sample_summary.csv`.

For OU, threshold-OU, MOU, transformed-OU, and semiparametric copula models the state is the log price

$$X_t = \log P_t, \qquad X_t = 0 \iff P_t = 1. \tag{1}$$

Price CIR and MCIR instead use $P_t > 0$ directly. Depeg-pressure CIR uses $|P_t - 1| + 10^{-8}$ and is interpreted as an amplitude model, not a signed mean-reversion model. Transition pairs are formed by timestamp hashing: a pair (X_t, X_{t+h}) is used only when the timestamp difference is exactly h. Missing bars are never interpolated.

3 Empirical specifications and estimators

3.1 OU benchmark and finite-horizon drift

The OU benchmark is

$$\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0, \quad \sigma > 0. \tag{2}$$

For an interval h, its conditional mean and variance are

$$m_h(x) = \theta + e^{-\kappa h}(x - \theta), \tag{3}$$

$$v_h = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa h}\right). \tag{4}$$

OU0 fixes $\theta = 0$, whereas OUF estimates θ. Parameters are estimated by exact Gaussian transition maximum likelihood. Ordinary Hessian standard errors, UTC-day cluster-sandwich standard errors, and UTC-day block bootstrap intervals are reported.

The model-free diagnostic estimates

$$\mu_h(x) = \frac{1}{h}\mathbb{E}[X_{t+h} - X_t \mid X_t = x] \tag{5}$$

by local-linear regression on a 99-point quantile grid. Bandwidth is selected by blocked cross-validation and uncertainty is computed from 1,000 UTC-day bootstrap replications. The sign-consistency ratio is

$$\mathrm{SCR}_h(\theta) = \frac{1}{G}\sum_{g=1}^{G} \mathbf{1}\{(\theta - x_g)\widehat{\mu}_h(x_g) > 0\}\,. \tag{6}$$

3.2 Parametric mixed-copula transition kernel

Let $P_h(x, \mathrm{d}y)$ be the transition kernel of a stationary base process with invariant distribution F. The article's constant-intensity Poisson construction gives

$$Q_h(x, \mathrm{d}y) = e^{-\lambda h} P_h(x, \mathrm{d}y) + \left(1 - e^{-\lambda h}\right) F(\mathrm{d}y). \tag{7}$$

For an OU base process, the conditional density used in estimation is therefore

$$q_h(y \mid x) = e^{-\lambda h}\phi(y; m_h(x), v_h) + \left(1 - e^{-\lambda h}\right)\phi\left(y; \theta, \frac{\sigma^2}{2\kappa}\right). \tag{8}$$

This is exactly the transition density implied by the article's process $Y_t = Y_t^{(N_t)}$ when the component processes are stationary OU processes. It nests the OU at $\lambda = 0$ and satisfies

$$\mathbb{E}[Y_{t+h} - \theta \mid Y_t = x] = e^{-(\kappa+\lambda)h}(x - \theta). \tag{9}$$

Hence the base diffusion half-life is $\log(2)/\kappa$, while the conditional mean half-life of the mixed process is $\log(2)/(\kappa + \lambda)$. These two quantities must not be conflated.

MOU0 fixes $\theta = 0$; MOUF jointly estimates $(\theta, \kappa, \sigma, \lambda)$. MCIR replaces the OU transition and Normal invariant density in Equation (8) with the CIR transition and its Gamma invariant density. Because the price-CIR likelihood is weakly identified, MCIR holds the base CIR parameters fixed and estimates only λ; MCIR is therefore a restricted diagnostic, not a full joint MLE.

3.3 Semiparametric multi-horizon copula

Let F_n be the empirical marginal CDF and $U_t = F_n(X_t)$. For a Gaussian base copula with $\rho_h = e^{-\kappa h}$, the fitted copula density is

$$c_h^{\mathrm{mix}}(u, v) = e^{-\lambda h} c_{\mathrm{G}}(u, v; \rho_h) + 1 - e^{-\lambda h}. \tag{10}$$

The parameters are estimated by an equally weighted composite likelihood over 5, 10, 30, 60, 120, and 360 minutes. Equation (10) defines a consistent stationary Markov copula construction. It does *not*, for an arbitrary F_n, automatically satisfy the sign restriction in the article's proposition for Gaussian copulas with general stationary marginals. This restriction is evaluated after estimation rather than assumed.

To test whether the failure at long horizons is caused by the single exponential correlation curve, define the two-scale correlation

$$\rho_h^{(2)} = w\exp(-\kappa_f h) + (1 - w)\exp(-\kappa_s h), \qquad 0 < w < 1, \quad 0 < \kappa_s < \kappa_f, \tag{11}$$

and the two-scale mixed density

$$c_h^{(2,\mathrm{mix})}(u, v) = e^{-\lambda h} c_{\mathrm{G}}(u, v; \rho_h^{(2)}) + 1 - e^{-\lambda h}. \tag{12}$$

Equation (11) is generated by the sum of two independent stationary OU factors. The latent vector is Markov, but its scalar sum is in general not one-dimensional Markov. Consequently, Equations (11)–(12) are used to diagnose multi-horizon dependence and density forecasts; they are not presented as a new instance of the article's one-dimensional Markov theorem. Four models are fitted on exactly the same deterministic subsample at each horizon: SingleGaussian, SingleMixed, TwoScaleGaussian, and TwoScaleMixed.

3.4 Prespecified model hierarchy

The hierarchy is fixed before interpreting the additional results. OUF and MOUF are the *core structural comparison*, because their transition laws directly match the article. Heteroskedastic threshold OU is the *main empirical extension* for conditional asymmetry. The four empirical-marginal copula models are *diagnostic extensions*. Random walk and nested-OU forecasts are *prediction benchmarks*. CIR, MCIR, and affine transformed models are *appendix feasibility checks*. A diagnostic model can reveal a missing dependence scale without inheriting the theoretical status of the core models.

Source file: `output/tables/article_model_hierarchy.csv`.

3.5 Model comparison and inference

Likelihood, AIC, and BIC are compared only for models fitted to the same state, sample, and likelihood. PIT uniformity and Ljung–Box tests assess transition density calibration. Because $\lambda = 0$ is a boundary point, the ordinary χ^2 likelihood-ratio reference distribution is not used. The primary parametric bootstrap uses the same deterministic 25,000 pairs for the observed and simulated statistics, holds the observed regressors fixed under the null, and jointly re-estimates $(\theta, \kappa, \sigma, \lambda)$ under MOUF in every replication. Out-of-sample comparison uses negative log score (NLS), CRPS, MAE, RMSE, Brier score, event log score, AUC, and calibration. All 2025 predictions use parameters selected without 2025 outcomes.

4 Empirical results

4.1 Nonparametric evidence and OU benchmark

Table 3: Finite-horizon mean-reversion sign diagnostics

Anchor definition	$\widehat{\theta}$	SCR	95% bootstrap interval	Grid points
Peg anchor	0	0.7778	[0.7677, 0.8788]	99
Sample anchor	6.700×10^{-5}	0.8788	[0.8586, 0.9697]	99

Source file: `output/tables/sign_test_baseline.csv`.

The point estimates predominantly have the correct direction, but the simultaneous confidence band in Figure 1 contains zero over a large part of the state grid. The evidence is therefore directional rather than a proof of pointwise significant infinitesimal mean reversion.

The heteroskedastic threshold OU estimates $(\kappa_+, \kappa_-) = (1.5674, 1.8103)$ and $(\sigma_+, \sigma_-) = (0.001248, 0.001515)$. Its likelihood gain is driven mainly by asymmetric conditional volatility rather than by a large difference in reversion speed.

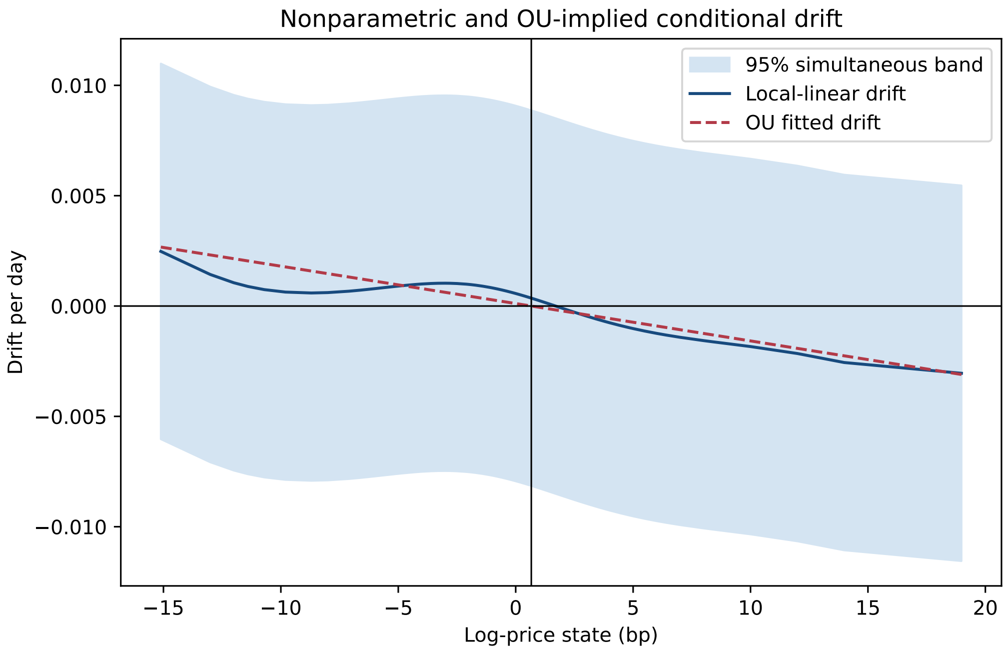


Figure 1: Local-linear finite-horizon drift and the OUF-implied drift.

Table 4: OU, asymmetric OU, and MOU parameter estimates

Model	$10^4\widehat{\theta}$	$\widehat{\kappa}$	$\widehat{\lambda}$	$\widehat{\sigma}$	Base HL (min)	Effective HL (min)
OUF	0.665	1.6932	—	0.001355	589.5	589.5
MOUF	7.248	0.02484	1.8863	0.000963	40,190.4	522.3

Rates are per day. OUF UTC-day cluster standard errors are $(0.166, 0.2099, 0.0001456)$ for $(10^4\theta, \kappa, \sigma)$. MOUF cluster standard errors are $(6.496, 0.01419, 0.18835, 0.00000950)$ for $(10^4\theta, \kappa, \lambda, \sigma)$. The large difference between the OUF and MOUF estimates of θ is a warning against interpreting the MOUF stationary mean as the economic peg without further stationarity restrictions.

Source file: `output/tables/model_estimates.csv`.

4.2 Does the mixed kernel add explanatory power?

Table 5: Comparable five-minute transition likelihoods

Model	Parameters	Log likelihood	AIC	BIC
OU0	2	4,207,015.02	$-8,414,026.04$	$-8,414,003.70$
OUF	3	4,207,021.25	$-8,414,036.50$	$-8,414,002.98$
Threshold OU, heteroskedastic	4	4,211,719.53	$-8,423,431.07$	$-8,423,386.38$
MOU0	3	4,351,515.34	$-8,703,024.68$	$-8,702,991.17$
MOUF	4	4,351,562.04	$-8,703,116.07$	$-8,703,071.39$

Source file: `output/tables/model_comparison.csv`.

Relative to OUF, MOUF increases the total log likelihood by 144,540.79, or approximately 0.2755 per transition pair, and lowers AIC by 289,079.58. This is the clearest in-sample evidence for the mixed kernel. The gain reflects a two-scale transition density: a highly persistent narrow base component and a small-probability draw from a much wider invariant component.

The primary boundary bootstrap removes the design mismatch in the earlier diagnostic. Both the observed and null-simulated statistics use the same deterministic systematic sample of 25,000 out of 524,610 exact transitions, the same fixed regressor values and five-minute step, the same OUF refit, and the same joint MOUF optimization. The observed statistic is $LR = 12{,}722.80$. Among $B = 500$ null replications, none is as large (maximum 0.635), giving the finite-replication correction

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{LR_b \geq LR_{\text{obs}}\}}{B+1} = \frac{1}{501} = 0.001996. \tag{13}$$

The alternative optimizer reports convergence in 98.2% of replications and $\widehat{\lambda}$ is at its zero boundary in 99.4%, as expected for this nonregular null. Thus the ordinary χ^2 approximation is inappropriate, but the matched conditional bootstrap provides strong evidence that the MOUF density adds a component absent from OUF on this prespecified design. The claim remains conditional on the 25,000-pair systematic design and the maintained OU null; it is not a proof that the economic data-generating process is MOUF.

Source file: `output/tables/matched_joint_mou_lr_bootstrap.csv`.

Source file: `output/tables/matched_joint_mou_lr_bootstrap_draws.csv`.

The former $B = 200$ profiled calculation is retained only as a historical sensitivity file because it compares a full-sample observed statistic with shorter simulated paths and does not jointly fit the same alternative. It is not used for the primary significance statement.

Source file: `output/tables/mixed_lr_bootstrap.csv`.

Three reported reset intensities answer different questions: These estimates are not interchange-

Table 6: Reset-intensity estimates under different objectives

Estimation objective	$\widehat{\lambda}$ per day	Interpretation
Joint five-minute MOUF likelihood	1.8863	Jointly changes the OU base and reset component
Matched joint MOUF likelihood	1.9185	Same 25,000-pair design as the boundary bootstrap
Restricted boundary profile	0.7678	Profiles λ conditional on a re-estimated OU base
Multi-horizon empirical-marginal copula	0.2557	Composite dependence fit over 5–360 minutes

able. Their dispersion is direct evidence that κ and λ are sensitive to the marginal model, estimation objective, and horizon set.

4.3 Multi-horizon dependence

Table 7: Empirical and fitted Gaussian-rank dependence

h (min)	Pairs	Empirical	Base fit	Mixed fit	Reset weight
5	524,610	0.99128	0.99692	0.99603	0.00089
10	524,494	0.98868	0.99384	0.99208	0.00177
30	524,279	0.98067	0.98165	0.97643	0.00531
60	524,143	0.97160	0.96363	0.95342	0.01060
120	524,017	0.95664	0.92858	0.90901	0.02108
360	523,734	0.91630	0.80068	0.75111	0.06192

Source file: `output/tables/copula_horizon_fit.csv`.

The mixed term reduces absolute dependence error at 5 and 10 minutes, but makes the fit worse from 30 minutes onward. At 360 minutes the data remain much more persistent than either single-exponential specification. A constant-intensity independence reset therefore does not replace a slow latent factor, a regime switching model, or a multi-scale OU structure.

The prespecified two-scale comparison confirms that the missing persistence is a separate dimension from the reset mixture.

Table 8: Single- versus two-scale empirical-marginal copula fits

Model	$\widehat{\kappa}_s$	$\widehat{\kappa}_f$	$\widehat{w}$	$\widehat{\lambda}$	Mean log score	Dependence RMSE
SingleGaussian	1.1398	—	—	0	1.42051	0.07022
SingleMixed	0.8889	—	—	0.2505	1.46991	0.07018
TwoScaleGaussian	0.3348	251.0089	0.01244	0	1.52663	0.00377
TwoScaleMixed	0.1361	225.0237	0.01215	0.5078	1.59503	0.03223

Rates are per day. "Mean log score" is the equally weighted mean copula log density over six horizons, using at most 100,000 deterministic pairs per horizon; higher is better. Dependence RMSE compares the six empirical Gaussian-rank correlations with the fitted correlations; lower is better. All four optimizers converged, and repeated starts attained the same best solution.

Source file: `output/tables/copula_scale_model_comparison.csv`.

The two metrics answer different questions. TwoScaleGaussian reduces dependence RMSE by 94.6% relative to SingleGaussian and reproduces the 360-minute dependence as 0.9083 versus the empirical 0.9163. TwoScaleMixed has the highest full-density score, improving it by 0.0684 over TwoScaleGaussian, but its reset component lowers the fitted 360-minute dependence to 0.8410 and therefore increases correlation RMSE. Thus the slow factor explains the long-memory shape, whereas the mixed component explains additional features of the joint density; a likelihood gain alone is not evidence of a better correlation curve.

Source file: `output/tables/copula_scale_horizon_fit.csv`.

For the empirical marginal F_n, the fitted semiparametric conditional mean has five-minute sign consistency of only 34.34% around the exact peg and 42.42% around the empirical marginal mean 6.713×10^{-5}. Thus the empirical- marginal Gaussian/mixed-copula fit is useful as a dependence diagnostic, but it is not empirically verified as a mean-reverting process under the article's definition. This conclusion is distinct from the parametric MOU result, whose Normal invariant marginal and OU base satisfy the theoretical construction by design.

Source file: `output/tables/copula_conditional_drift_baseline.csv`.

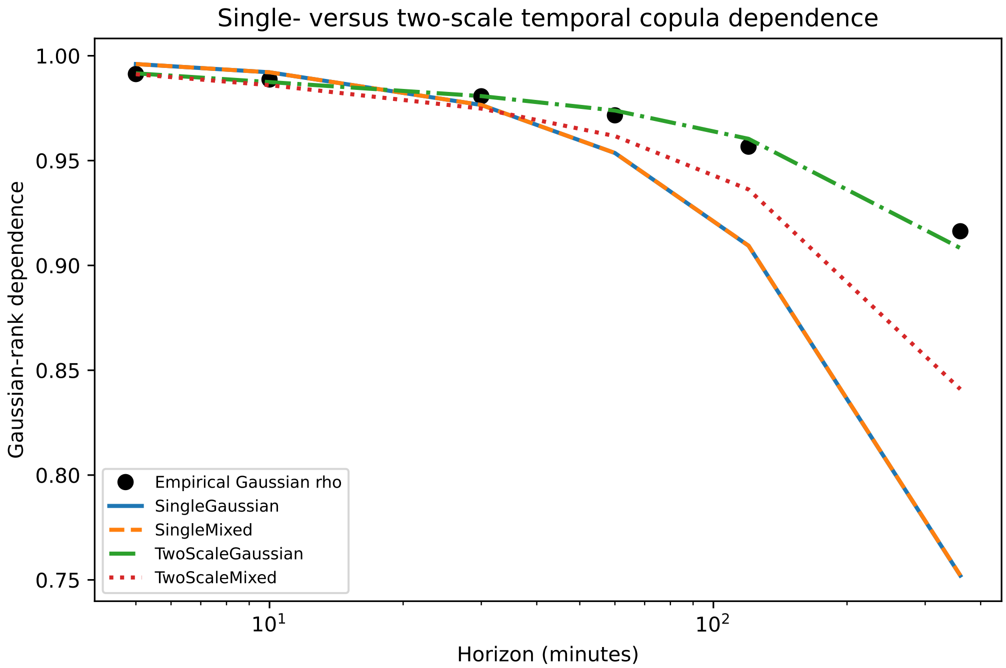
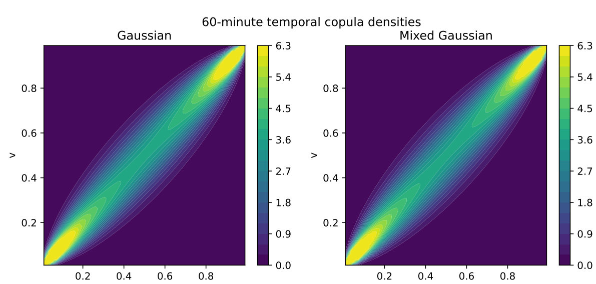
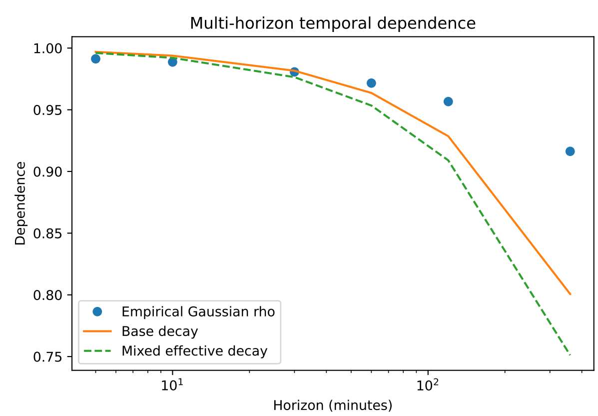


Figure 2: Empirical multi-horizon dependence and the four scale/reset specifications.



(a) Gaussian and mixed copula contours.



(b) Dependence across horizons.

Figure 3: Copula shape and multi-horizon dependence.

Source file: `output/tables/descriptive_statistics_baseline.csv`.

4.4 Density diagnostics and out-of-sample performance

Numerical optimization succeeds for OUF and MOUF, but both models are rejected by PIT uniformity and by Ljung–Box tests of transformed residuals and squared residuals (reported numerical p-values are effectively zero). The mixed model therefore improves relative likelihood without achieving absolute transition- density adequacy.

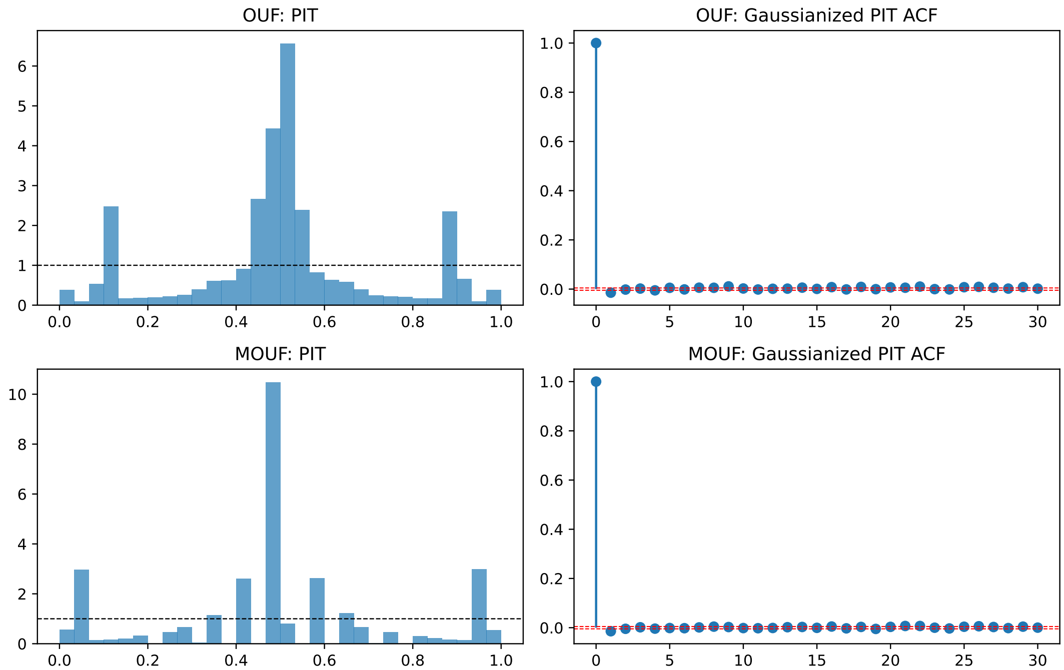


Figure 4: PIT and residual diagnostics for OUF and MOUF.

Table 9: Validation-period model selection, 2024

Model	Mean NLS	Mean CRPS	Mean RMSE	NLS rank
OUF	−7.18759	0.00013091	0.00023667	1
Random walk	−7.13145	0.00013279	0.00021416	2
Nested-OU proxy	−7.13006	0.00014465	0.00026387	3
MOUF	−7.10424	0.00015907	0.00029675	4

Source file: `output/tables/validation_model_selection.csv`.

Lower NLS and CRPS are better. OUF is selected on the validation period. In the 2025 test period, MOUF has a better 5-minute NLS for the 50-basis-point depeg specification (−8.7632 versus −8.7025 for OUF), but OUF is better at 60, 360, and 1,440 minutes. The mixed kernel therefore has a localized short-horizon density advantage rather than uniform out-of-sample dominance. *Source file:* `output/tables/forecast_scores.csv`.

The scale/reset comparison is a separate copula-density exercise. The empirical marginal and all copula parameters are estimated using 2021–2023 only; the fitted objects are then frozen for 2024 and 2025. Table 10 reports the equally weighted mean across the six horizons.

Table 10: Out-of-sample copula log scores by model scale

Model	Training 2021–2023	Validation 2024	Test 2025
SingleGaussian	1.27385	1.54637	1.38939
SingleMixed	1.31164	1.59861	1.43596
TwoScaleGaussian	1.40784	1.61073	1.50638
TwoScaleMixed	1.47899	1.66640	1.57116

Higher log score is better. Each period-specific mean first averages over observations within a horizon and then gives equal weight to each of the six horizons. No validation or test outcomes enter parameter estimation.

Source file: `output/tables/copula_scale_oos_scores.csv`.

TwoScaleMixed exceeds SingleMixed by 0.06780 log-score units in validation and 0.13520 in test; it also exceeds TwoScaleGaussian by 0.05567 and 0.06478. This is the clearest out-of-sample metric in favor of combining a slow factor with the mixed reset. The advantage is not uniform by horizon: on the 2025 test data the two-scale models score worse than SingleMixed at 5 and 10 minutes, but better at 30, 60, 120, and 360 minutes. The result supports multi-scale dependence, not an unqualified claim that the mixed process dominates every forecast horizon.

The baseline 50/25-basis-point event rule identifies eight events. Their empirical recovery-time median is 155 minutes, compared with 665 minutes under OUF simulation and 1,085 minutes under MOUF simulation. The reset mechanism does not necessarily move the state toward the peg; it draws from the invariant marginal. Consequently, the effective conditional-mean half-life in Equation (9) is not a first-passage-time statistic.

Source file: `output/tables/first_passage_comparison.csv`.

4.5 Frequency, sample, and block-length robustness

Table 11: Baseline frequency robustness

Minutes	Exact pairs	$\widehat{\kappa}$	$\widehat{\lambda}$	OU HL	Effective HL
5	524,610	1.6932	0.6969	589.5	417.6
15	174,578	1.8504	0.1784	539.4	492.0
30	87,068	1.3992	0.1323	713.4	651.7
60	43,322	1.1311	0.0785	882.5	825.2

All non-five-minute rows use complete aggregation bins only. Half-lives are in minutes. In this table λ is estimated conditionally on fast OU parameters, so it should not be compared numerically with the joint MOUF estimate without qualification.

Source file: `output/tables/robustness_estimates.csv`.

The decline in $\widehat{\kappa}$ at lower sampling frequencies is consistent with microstructure noise inflating high-frequency reversion estimates. The 1-, 3-, and 7-day moving-block bootstrap standard errors for OUF κ are 0.2021, 0.2300, and 0.2622, respectively; uncertainty increases with block length but the reversion-rate estimate remains positive. In the full 2017–2025 sample, retaining three flagged hard anomalies produces $\widehat{\kappa} = 303.09$, whereas excluding them gives $\widehat{\kappa} = 2.742$. Structural interpretation of the full-sample high-frequency estimate is therefore untenable without explicit outlier treatment.

Source file: `output/tables/ou_block_length_robustness.csv`.

5 Full-model registry and validation selection

The second-stage exercise removes the MVP complexity screen. Every article model with an explicit finite-dimensional observation density, and every benchmark or marginal–copula family required by the empirical methodology, is either estimated or assigned a terminal failure diagnostic before any model is chosen for emphasis. The registry contains 25 theory/methodology objects. Of these, 18 are estimated in the current or pre-existing pipeline, five terminate with a retained monotonicity, support, or endpoint failure, the stationary Gaussian–Gamma construction is estimable but fails the article's shape condition, and the bond-pricing PDE is retained as not observable from spot data.

Source file: `output/tables/full_model_implementation_status.csv`.

For every comparable model, selection uses the same 2024 five-minute conditional log score on the log-price measure,

$$\widehat{S}_m^V = \frac{1}{N_V}\sum_{i=1}^{N_V} \log q_m(x_{t_i} \mid \mathcal{F}_{t_i-}), \qquad N_V = 40{,}000. \tag{14}$$

Price-density models are converted by the Jacobian $q_X(x_t \mid x_{t-h}) = q_P(e^{x_t} \mid e^{x_{t-h}})e^{x_t}$. Copula models use $q_X(x_t \mid x_{t-h}) = c_h(F(x_{t-h}), F(x_t))f(x_t)$. Thus AIC values from ordinary and composite likelihoods do not enter this cross-class ranking. For the two-state OU, the primary one-step score is the exact hidden-state predictive density

$$q_t(x_t \mid \mathcal{F}_{t-1}) = \sum_{j=1}^{2}\left(\sum_{i=1}^{2} \alpha_{t-1,i} p_{ij}\right) p_j(x_t \mid x_{t-1}), \tag{15}$$

with the filter reset to its stationary probabilities after every timestamp gap. Longer-horizon Markov-switching entries are labelled as endpoint-mixture approximations and do not determine selection.

Table 12: Unified 2024 validation ranking at the five-minute horizon

Rank	Model	Mean log score	Coverage	Identification audit
1	Markov-switching OU	8.824686	1.000	weak: slow regime/boundary
2	Student-t margin + Student-t copula	8.781202	1.000	weak: $\nu = 2.05$ grid boundary
3	Student-t margin + mixed Student-t copula	8.781201	1.000	weak: $\lambda \simeq 0$, ν boundary
23	MOUF	8.431420	1.000	passes numerical identification gate
43	OUF	8.221318	1.000	passes numerical identification gate
44	Random walk	8.220764	1.000	no-reversion benchmark

The rank numbers for MOUF, OUF, and the random walk are their positions in the complete candidate table; omitted rows are not omitted from estimation or ranking. Selection uses 2024 only. The 2025 scores are post-extension exploratory because aggregate 2025 outcomes had been inspected previously.

Source file: `output/tables/full_model_validation_ranking.csv`.

The selected model's advantage is not merely a point-ranking artifact. Day-block resampling of paired score differences gives the comparisons in Table 13. These intervals measure predictive-score uncertainty conditional on the frozen estimates; they do not repair weak identification of the switching parameters.

The expanded grid also makes the mixed-copula claim more precise. With the Student-t marginal fixed, adding the independent-reset component raises the Gaussian validation score by

Table 13: Paired day-block comparison of the selected model

Reference	$\widehat{S}_{MSOU} - \widehat{S}_{ref}$	95% block interval	Daily win share
Student-t margin + Student-t copula	0.043484	[0.018242, 0.065954]	0.656
MOUF	0.393266	[0.355106, 0.429977]	0.915
OUF	0.603368	[0.566379, 0.640416]	0.956
Random walk	0.603922	[0.564066, 0.642485]	0.959

Source file: `output/tables/full_model_pairwise_validation.csv`.

0.078347 and the two-scale Gaussian score by 0.048432. It changes the Student-t copula score by only -6.95×10^{-7} because $\widehat{\lambda}$ is effectively zero. For Frank, Clayton, survival Clayton, Gumbel, and survival Gumbel, the fitted mixed versions hit weak-identification/boundary diagnostics and reduce the validation score by 1.98 to 3.21. The empirical advantage is therefore specific to the Gaussian and two-scale Gaussian reset constructions; it is not a theorem that mixing improves every copula family.
Source file: `output/tables/full_mixture_incremental_scores.csv`.

The transformation audit is equally informative. Affine OU is estimable; exponential OU converges but is weakly identified. The OU special-function solution is even in the latent state and hence non-injective on $\mathbb{R}$. Affine CIR is estimated but weak, while quadratic CIR, exponential CIR, and the prespecified CIR special-function branches fail global monotonicity or observed support. These failures are substantive outcomes, not silent exclusions. The Gamma Case 1 equation has no finite shape root on the audited range, so Case 2 lacks the stationary endpoint it requires. Finally, the mixed-copula bond PDE cannot be estimated without a bond-price cross section and a risk-neutral measure; its equation is implemented only as a non-observable theoretical object in this spot-price application.

6 What the evidence establishes

The empirical evidence supports the following ordered conclusions.

1. USDT/USD log prices display finite-horizon directional reversion around a near-zero log-price anchor. A simple OU provides a transparent first benchmark, but its transition density is misspecified.

2. The article's parametric mixed-copula construction is implemented exactly for stationary OU and CIR bases. For the OU base it yields a very large in-sample likelihood gain. The same-design, jointly re-estimated boundary bootstrap rejects the OU restriction at the minimum attainable corrected level 1/501, conditional on the prespecified 25,000-pair design.

3. The gain should be interpreted as evidence for a two-component transition density, not as causal evidence that λ measures redemption, arbitrage, news, or any unique economic event.

4. MOUF does not dominate OUF in the original validation exercise, PIT diagnostics, or empirical first-passage times. A constant independent-reset intensity is therefore an incomplete empirical description of the price process.

5. Adding a slow latent factor reduces six-horizon dependence RMSE from 0.07022 to 0.00377. Adding the reset mixture to the two-scale model raises validation and test copula log scores

to 1.66640 and 1.57116, the best of the four diagnostic models, but worsens correlation RMSE to 0.03223. Persistence shape and density fit are distinct empirical targets.

6. The empirical-marginal semiparametric copula is not shown to satisfy the article's sufficient mean-reversion condition. It must be labelled as a dependence and forecasting extension unless the marginal sign restriction is imposed during estimation.

7. All article transformation families with explicit finite-dimensional forms have now been implemented. Their failures matter empirically: nonlinear CIR variants fail monotonicity/support, exponential OU is weakly identified, and the OU special-function map is non-injective. The affine-OU calibration is the only transformation result suitable for routine comparison.

8. Complexity-blind 2024 validation selects the two-state Markov-switching OU on predictive log score, not a mixed-copula model. Its paired advantage over the best Student-t copula model is positive, but its near-zero slow-regime reversion rate is weakly identified. It may be emphasized as a predictive benchmark, while structural interpretation of its regimes must remain qualified.

The most defensible statement of the mixed-copula advantage is therefore: *relative to an OU transition, the Poisson-switching mixed kernel captures a narrow persistent component and rare wide transitions, produces a large matched- design in-sample likelihood gain, and preserves the base invariant marginal and Markov semigroup. Within the diagnostic two-scale family, the mixed reset also improves validation and test copula log scores for Gaussian and two-scale Gaussian dependence.* Its limitation is equally clear: the reset does not by itself generate the observed slow persistence, is redundant for the fitted Student-t copula, fails out of sample for the fitted Archimedean mixed variants, worsens the two-scale correlation RMSE, and does not reproduce empirical recovery times. It remains the article's main theoretical contribution, but the empirical headline model is chosen by the frozen validation rule rather than by ownership or complexity.

A Theory and implementation audit checklist

- ✓ MOU density equals $e^{-\lambda h}p_{\mathrm{OU}} + (1 - e^{-\lambda h})f_{\mathrm{OU,stat}}$ pointwise. The code test verifies exact equality to OU when $\lambda = 0$.

- ✓ MCIR uses the stationary Gamma law implied by the same (θ, κ, σ) as the CIR base.

- ✓ Mixed conditional mean decays at $\kappa + \lambda$; simulation and analytic moments are compared in the test suite and generator-validation table.

- ✓ Affine OU and CIR inverse transformations and Jacobians match the article's linear formulas under $A_0 = 1$ and $B_0 = 0$.

- ✓ All primary five-minute pairs are exact timestamp matches; no missing bar is treated as an observed transition.

- △ The article assumes a stationary base process. Rolling estimates show substantial parameter variation, so stationarity is an empirical approximation rather than an established fact.

- △ The semiparametric empirical marginal does not impose the article's Gaussian-copula marginal sign restriction and fails the finite-horizon sign diagnostic on most grid points.

- △ Price-CIR likelihood evaluation requires a high-noncentrality Gaussian moment fallback for all evaluated price transitions; price CIR and affine transformed CIR are flagged as weakly identified.
- ✓ The primary boundary bootstrap uses the same 25,000 transition pairs and the same joint OUF/MOUF estimators for the observed and all 500 simulated statistics. Its inference is conditional on the systematic design and fixed observed regressors, rather than an unconditional full-sample statement.
- △ The two-scale scalar copula model captures multi-horizon dependence but is not one-dimensional Markov; it is a diagnostic latent-factor extension rather than an implementation of the article's scalar Markov theorem.
- ✓ Deterministic seasonal $c(t)$, exponential OU, quadratic and exponential CIR, and the article's special-function branches are implemented. Converged, weak, non-monotone, and support-failed outcomes are all retained in the same implementation-status table.
- ✓ Normal, Student-t, skew Student-t, GED, and monotone-spline margins are crossed with Gaussian, Student-t, Frank, Clayton, survival Clayton, Gumbel, and survival Gumbel copulas; base and independence-mixed versions are scored for all three periods and four horizons.
- ✓ Random-walk, jump-OU, nested-OU, two-state OU, and a finite-dimensional deterministic time-varying marginal–copula model enter the same primary validation ranking. No model is removed because it is complex.
- △ Markov-switching OU wins the exact filtered one-step score but its slow-regime rate is near the optimization boundary and spans substantially less than one mean-reversion unit in the continuous estimation segment.
- × The mixed-copula bond-pricing PDE is not empirically observable from a USDT spot-price series without a bond-price cross section and a specified risk-neutral measure. This is retained as an explicit data-identification failure rather than an unimplemented model.

B Output provenance

Table 14: Primary reproducibility files

File	Content
output/tables/data_audit.csv	Raw-data hash, rows, timestamps, gaps, and audit conditions
output/tables/sample_summary.csv	Sample periods, coverage, and exact-pair counts
output/tables/sign_test_baseline.csv	Nonparametric sign-consistency statistics
output/tables/model_estimates.csv	Parameters and ordinary, cluster, and bootstrap uncertainty
output/tables/model_comparison.csv	Likelihood, information criteria, numerical checks, and PIT tests
output/tables/matched_joint_mou_lr_bootstrap.csv	Primary matched-design joint-MOUF boundary bootstrap
output/tables/matched_joint_mou_lr_bootstrap_draws.csv	All 500 null LR draws and convergence diagnostics
output/tables/mixed_lr_bootstrap.csv	Superseded restricted boundary-bootstrap sensitivity
output/tables/copula_horizon_fit.csv	Empirical and fitted dependence over six horizons
output/tables/copula_scale_model_comparison.csv	Single/two-scale fit, parameters, and dependence RMSE
output/tables/copula_scale_horizon_fit.csv	Four-model fitted dependence at each horizon
output/tables/copula_scale_oos_scores.csv	Frozen-training-model validation and test copula log scores
output/tables/article_model_hierarchy.csv	Structural, extension, diagnostic, benchmark, and appendix tiers
output/tables/copula_conditional_drift_baseline.csv	Semiparametric conditional means and sign diagnostic
output/tables/forecast_scores.csv	Validation and test forecast scores
output/tables/first_passage_comparison.csv	Empirical and simulated recovery-time distributions
output/tables/robustness_estimates.csv	Frequency, extended-sample, and anomaly-policy robustness
output/tables/full_model_registry.csv	Frozen article/methodology model universe and comparison groups
output/tables/full_model_implementation_status.csv	Convergence, weak identification, monotonicity, support, and retained failures
output/tables/full_model_validation_ranking.csv	Complexity-blind 2024 five-minute conditional-density ranking
output/tables/full_model_pairwise_validation.csv	Paired day-block uncertainty for the selected model
output/tables/full_mixture_incremental_scores.csv	Within-family mixed-minus-base score increments
output/tables/updated_stage_gate.csv	Full-model coverage, scoring, diagnostics, and no-leakage gates
output/models/mixed_fits_baseline.json	Full MOU/MCIR parameter and optimization records
output/models/full_model_fits.json	Full-suite fits and nonlinear failure diagnostics
output/models/full_copula_fits.json	Five margins and fourteen fitted copula specifications
output/models/semiparametric_copula_baseline.json	Multi-horizon composite-likelihood estimate
output/models/copula_scale_models.json	Full-sample and training-sample scale/reset estimates
output/run_status.json	Step-level one-click pipeline status
output/manifest.json	Artifact hashes and joint test/pipeline completion status